Assignment 2

Due date: October 16, 2026

Midterm: October 22, 2026

Q1. Compute (4) and (5) in Proposition 1 of Lecture 4.

Q2. The total equity value of the firm is given by

$$\begin{aligned} &S(V;V_B) = v(V;V_B) - D(V;V_B) \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi} \\ &\quad - P\frac{\rho+m}{r+m} + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right](V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}. \end{aligned}$$

Show that the solution of

$$\frac{\partial S(V;V_B^*)}{\partial V}|_{V=V_B^*} = 0$$

is given by $V_B^* = \epsilon P$, where

$$\epsilon = \frac{\frac{\rho+m}{r+m}\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}.$$

A Laplace Transforms of First Passage Times

Consider the first passage time

$$\tau_b = \min\{t \geq 0 :\ W_{\mu,\sigma}(t) = b\},$$

where $W_{\mu,\sigma}(t)$ is a Brownian motion with drift μ and volatility σ, i.e. $W_{\mu,\sigma}(t) = \mu t + \sigma W(t)$. Note that $W_{\mu,\sigma}(\tau_b) = b$ on the set $\tau_b < \infty$.

For any $s > 0$, the Laplace transform of τ_b is given by

$$\mathbb{E}[e^{-s\tau_b}] = \begin{cases} \exp\left\{-\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right)\right\}, & \text{if } b > 0, \\ \exp\left\{\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right\}, & \text{if } b < 0. \end{cases} \tag{A.1}$$